

Replicating Sadka (2010, JFE) Table III on current WRDS/LSEG TASS data

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1. Objective and framing

I set out to replicate Table III of Sadka (2010, JFE) — the time-series regressions of 11 equal-weighted hedge-fund style portfolios on the Fung–Hsieh factors and the Sadka liquidity factor — using the current WRDS/LSEG TASS vintage, with the aim of building a clean style-level baseline that could later be extended with additional factors. I report the paper-faithful keep-all version alongside two documented data-error screens; the main closest-feasible version uses the conservative screen, which removes only clearly impossible modern-vintage return records while preserving extreme returns of the same order of magnitude as those reported by Sadka. I would not describe this as an exact reproduction of Sadka’s original 2009-era TASS sample. The main qualitative pattern is close, but the sample counts and several coefficient magnitudes differ, and I have tried to be explicit about why.

2. Data and specification

Monthly TASS returns, Jan 1994–Dec 2008; Live + Graveyard; monthly, net-of-fees, USD, excess of the risk-free rate; 11 self-classified styles; equal-weighted monthly style portfolios; OLS regressions on (M1) MKT-RF + Liquidity and (M2) the full Fung–Hsieh seven factors + Liquidity. The rolling-beta / decile rules of Table IV were deliberately kept out.

3. Step-by-step: inputs and outputs

(a) Liquidity factor. *Input:* the WRDS Sadka file (permanent-variable component), used as-is. *Output:* the crisis months are all negative, in line with Figure 1 — 1998-09 = -0.016 , 2001-01 = -0.012 , 2007-08 = -0.012 , 2008-09 = -0.031 .

(b) Fung–Hsieh factors + Table II. *Input:* Ken French (MKT-RF, SMB), David Hsieh (PTFS), FRED (GS10, BAA for $\Delta\text{TERM}/\Delta\text{CREDIT}$). *Output:* the 28 pairwise correlations all fall within 0.05 of the paper, with matching signs. The key liquidity row:

Liquidity $\times$	MKT-RF	SMB	ΔTERM	ΔCREDIT	PTFSBD	PTFSFX	PTFSCOM
Sadka	0.13	0.07	0.08	-0.34	-0.01	-0.10	-0.08
Replication	0.14	0.08	0.09	-0.39	-0.05	-0.13	-0.09

(c) TASS sample. *Input:* the current LSEG schema (`tr_tass`): `productperformance` (returns) and `productdetails` (style, currency, gross/net, live/graveyard). *Output (filter funnel):* 840,794 returns / 15,980 funds $\rightarrow$ monthly 14,549 $\rightarrow$ USD 8,700 $\rightarrow$ net 8,638 $\rightarrow$ 11-style 8,345 funds (501k fund-months). All 11 styles present; Fund of Funds retained; Live and Graveyard both present.

4. Table III liquidity loadings: original vs. replication

Both models, each reported as the liquidity coefficient with $[t]$; the replication column uses the conservative data screen (S01, see §5).

Style	Model 1: $\beta_{LIQ} [t]$		Model 2: $\beta_{LIQ} [t]$	
	Sadka	Replication	Sadka	Replication
Convertible Arbitrage	1.28 [6.1]	1.39 [6.5]	0.78 [4.2]	0.76 [3.9]
Dedicated Short Bias	0.33 [0.9]	0.33 [1.0]	-0.16 [-0.5]	-0.14 [-0.5]
Emerging Markets	1.53 [3.3]	1.42 [3.1]	1.13 [2.4]	0.95 [2.0]
Equity Market Neutral	0.70 [5.2]	0.44 [3.3]	0.47 [3.6]	0.33 [2.3]
Event Driven	0.88 [5.7]	0.85 [4.7]	0.53 [4.1]	0.53 [3.0]
Fixed Income Arbitrage	0.94 [4.9]	0.68 [2.4]	0.39 [2.5]	0.34 [1.2]
Fund of Funds	1.09 [5.1]	0.65 [3.2]	0.74 [3.7]	0.39 [2.0]
Global Macro	0.42 [1.8]	0.15 [0.6]	0.38 [1.7]	0.17 [0.7]
Long/Short Equity	0.91 [3.8]	0.45 [2.1]	0.81 [3.7]	0.36 [2.0]
Managed Futures	0.34 [0.8]	0.20 [0.4]	0.39 [1.0]	0.30 [0.5]
Multi-Strategy	1.53 [5.1]	0.57 [4.1]	1.42 [4.9]	0.39 [2.7]

Interpretation. All 11 Model-2 liquidity-loading signs match the paper. Strictly using $|t| \geq 1.96$ in Model 2, six styles are clearly positive and significant (Convertible Arbitrage, Emerging Markets, Equity Market Neutral, Event Driven, Long/Short Equity, Multi-Strategy); Fund of Funds is very close to the cutoff, Fixed Income Arbitrage is positive but not significant, and Global Macro and Managed Futures remain weak — consistent with the paper. Dedicated Short Bias has a negative and insignificant liquidity loading and a market beta of -0.88 (Sadka -0.97). Several magnitudes (Multi-Strategy, Long/Short Equity, Fund of Funds) come out smaller than Sadka’s.

5. Sensitivity and caveats

Data-error screen. The raw current-vintage data contain a few extreme apparent data-error records (a $+31,208\%$ monthly return; exact 99.0 sentinels). I report three screens; the conservative one (S01) is the main version. It removes only 8 observations and does not delete returns of the same order of magnitude as Sadka’s published extreme values (post-screen maximum 18.16, compared with his published 13.245):

Screen	Funds	Max ret	Conv.Arb $\beta_{LIQ}(M2)$	FI.Arb $\beta_{LIQ}(M2)$	sign-match
S00 keep-all (sentinels only)	8,345	312.08	-1.43	-0.40	9/11
S01 conservative ($>2000\%/mo$)	8,345	18.16	0.76	0.34	11/11
S02 aggressive ($>100\%/mo$)	8,345	0.98	0.76	0.20	11/11

Sample counts. The Overall N is 8,345, versus Sadka’s 12,929. The gap is largely the USD filter: with all currencies the count is 13,521 ($\approx$ his total), but I keep USD-only because the paper specifies USD returns. **Managed Futures** is the largest coverage caveat — only about 17 funds carry that primary category in the current 1994–2008 LSEG vintage before the final USD/net/style filters, and only 3 remain in the final USD sample (Sadka had 964); fund names are anonymized, so I do not force-map “Other”/“Undefined” funds into it. A few style counts (replication vs Sadka): Fund of Funds 2,458/4,268; Long/Short 2,509/3,574; Multi-Strategy 631/1,177.

6. Summary

The main liquidity-risk pattern that Table III documents is present on this different data vintage, even though it is not an exact match: the magnitudes are softer and the sample is smaller. The qualitative contrasts hold, and the code is organized so that additional factors can be added to the same style-level regressions. Full numbers, all three screens, and the audit trail are in the accompanying workbook.